

## Part D AI Interview Pipeline Design

### 1. RecruitView Paper

<https://arxiv.org/pdf/2512.00450>

### 2. Persuasiv AI

<https://persuasiv.ai/en>

### 3. Internal Demo System

<https://lilac-blemish-ongoing.ngrok-free.dev/>

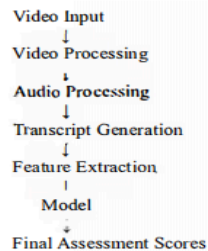
#### Task

1. Study the references and propose your own AI Interview Assessment Pipeline.
2. You are NOT required to replicate the existing systems exactly

3. You may improve, simplify, or redesign the workflow based on your understanding.

In the report, include:

1. Provide a complete flow diagram., eg.



\*You may include additional modules if necessary.

2. Explanation of each component in the pipeline
3. Compare your proposed system against the provided references:
  - a. RecruitView
  - b. Internal Demo System
  - c. Other reference found
4. Design an evaluation process to determine whether the AI Interview model is performing well.

Include:

Evaluation Metrics  
Validation Strategy  
Testing Procedure  
Human Evaluation Procedure  
Model Comparison Methodology

1.

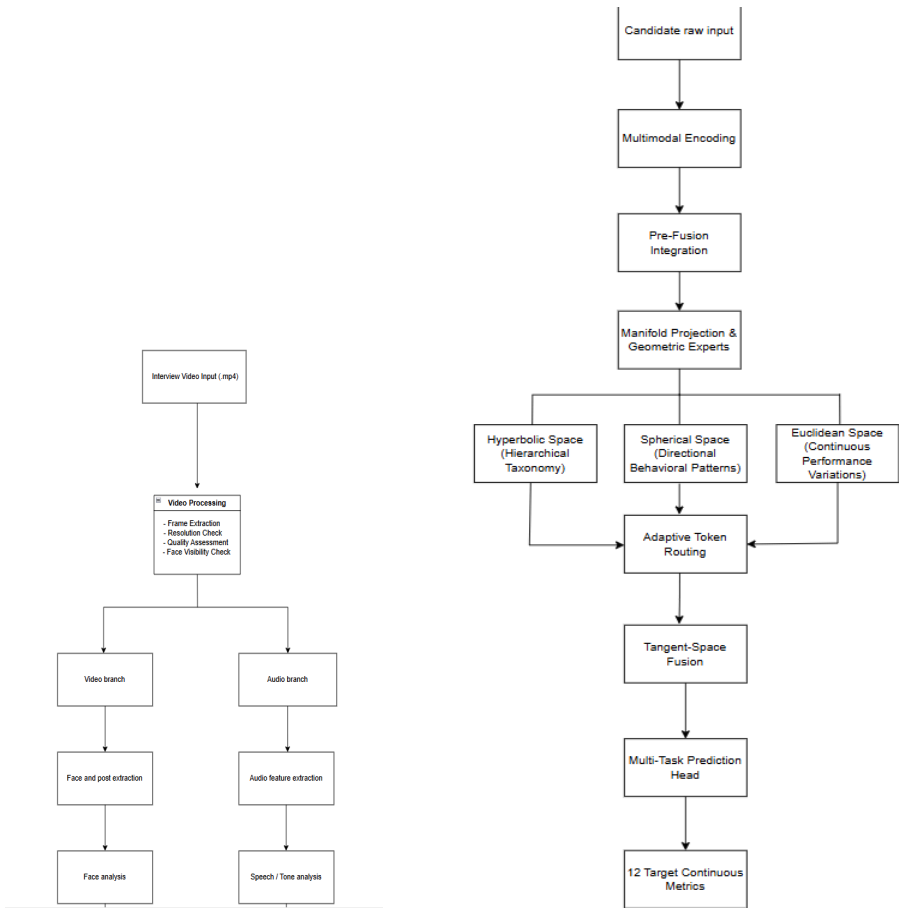


Figure 2

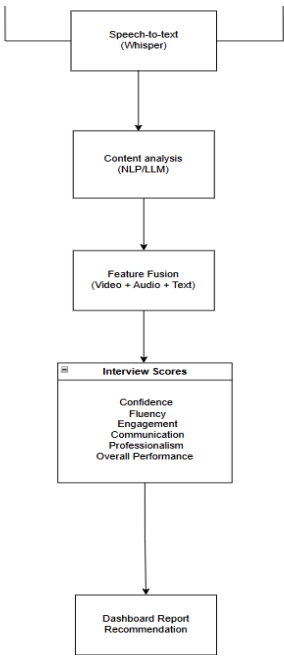


Figure 1

## 2. Explanation of Each Component (Figure 1)

- **Video input** - Input video that contains face, voice, spoken answers
- **Video processing** - missing transcripts, no face detected, multiple people, video too short, poor audio
- **Video processing** - use the OpenCV, MediaPipe and DeepFace libraries to extract the eye contact, emotion posture and facial engagement
- **Audio processing** - use the Librosa, Praat to extract the speech rate, pause ratio, pitch, voice energy and Jitter
- **Transcript generation** - use the whisper to output the candidate transcript
- **Content analysis** - use the Sentence-Bert and LLM to calculate the answer relevance, vocabulary richness and technical depth
- **Feature fusion** - combine the video features + audio features + text features into one feature vector
- **Assessment model** - use the XGBoost by input the feature vector to show the interview assessment
- **HR dashboard** - finalize all the scores and prompt out on the dashboard with the confidence score, fluency score, communication score and overall score.

## 2. Multimodal Feature Encoding Stage (Figure 2) \*after feature fusion

The system must capture signals distributed across what candidates say (linguistic), how they speak (acoustic), and how they present visually (visual).

- **Text/Linguistic Encoder**: Processes verbatim speech transcripts using a pre-trained language understanding model like **DeBERTa-v3-base** to output token representations.

### 4.3. Multimodal Encoding

We employ pretrained encoders for each modality: DeBERTa-v3-base [36] for text,

- **Audio/Acoustic Encoder**: Processes raw 16kHz audio waveforms utilizing self-supervised architectures like **Wav2Vec2** or **HuBERT** to extract acoustic and prosodic variations.

tience 5). Text is tokenized with max length 512, audio resampled to 16kHz, and video sampled at 16 FPS with 16 frames per clip resized to  $224 \times 224$ .

- **Visual/Temporal Encoder**: Captures facial expressions, posture, and body gestures. It maps cropped spatial video frames (e.g.,  $224 \times 224$  pixels at 16 FPS) via a visual backbone like **VideoMAE** or **TimeSformer**. It is highly recommended to append a temporal modeling pipeline (e.g., BiLSTM + Multi-Head Attention + 1D Convolution) to isolate fine-grained sequence dynamics.

frozen for parameter efficiency. For video, we apply a sophisticated temporal modeling pipeline consisting of BiLSTM, multi-head attention, and 1D convolution to capture rich temporal dynamics before pre-fusion. All

- Parameter Efficiency Note: For optimal resource utilization, freeze early encoder blocks and only fine-tune the final few transformer layers. Project all encoder outputs to a unified embedding space dimension (e.g. Figure 3).

## 4. Methodology

### 4.1. Problem Formulation

We address the problem of predicting multiple continuous attributes from multimodal behavioral data. Given a video clip containing visual frames  $\mathbf{V} \in \mathbb{R}^{T \times H \times W \times 3}$ , audio

dimension  $d_{model} = 768$ . Full details are provided in the

Figure 3

## 2. Pre-Fusion Module

- Concatenate the text, audio, and video feature sequences together.
- Add unique, learnable modality embeddings to preserve data origin identities across the combined sequence.
- Process the sequence through a multi-layer Transformer encoder to learn rich cross-modal contextual representations.
- Apply learned attention pooling (rather than a simple global average) to dynamically isolate and weight the frames or words most critical to human behavior assessment.

## 3. Manifold Projection & Geometry-Specific Experts

Traditional Euclidean networks often fail to preserve the geometric structure inherent to human traits. To capture complex psychological properties, project the pooled pre-fusion feature ( $\mathbf{z}_{pre}$ ) into three independent geometric manifolds:

- **Hyperbolic Space (Poincaré Ball Model)**: Best suited to map hierarchical taxonomies, such as major personality trait domains breaking down into specific localized facets.
- **Spherical Space (Unit Sphere Model)**: Explicitly handles directional behavioral data, such as shifting rotational gestures or facial muscle vector changes.
- **Euclidean Space (Standard Feature Plane)**: Efficiently models flat, un-structured continuous performance distributions.

Each manifold configuration requires a dedicated expert network to handle processing inside that specialized geometry (e.g., Möbius transformations in hyperbolic gyrovector space) followed by structural Geometry-Aware Intra-Manifold Attention mapped via local tangent-space conversions.

## 4. Adaptive Token Routing

Instead of utilizing fixed weights or hard-coded assumptions, deploy an Adaptive Token Router. This lightweight multi-layer perceptron (MLP) reads the input profile and dynamically generates routing weights ( $\mathbf{r} \in \Delta^{K-1}$ ) tailored specifically to that candidate's stylistic traits.

#### 4.7. Adaptive Token Routing

The router learns to weight expert outputs based on input characteristics. Given  $\mathbf{z}_{pre}$ , a lightweight MLP predicts routing weights:

$$\mathbf{r} = \text{softmax}(\mathbf{W}_r^{(2)} \sigma(\mathbf{W}_r^{(1)} \mathbf{z}_{pre} + \mathbf{b}_r^{(1)}) + \mathbf{b}_r^{(2)}) \quad (5)$$

- Regularization Requirement: Enforce a negative Entropy Regularization Loss ( $L_{entropy}$ ) alongside a Load Balancing Loss ( $L_{balance}$ ). This penalizes the model if it overly defaults to just one space, guaranteeing that all three geometric experts contribute their complementary views to the assessment.

perts. To encourage diverse geometry utilization, we apply entropy regularization:  $\mathcal{L}_{entropy} = -\lambda_{ent} \sum_{i=1}^K r_i \log r_i$ . A negative value encourages high entropy, promoting complementary geometric views.

#### 5. Tangent-Space Fusion Module

- To blend the outputs without running expensive iterative optimization patterns across incompatible spaces, map all hyperbolic and spherical expert vectors into a shared local tangent space using mathematical logarithmic maps.
- Compute a routing-weighted average of the Euclidean, spherical, and hyperbolic outputs within this shared tangent plane:

$$\mathbf{z}_{fusion} = r_h \mathbf{v}_h + r_s \mathbf{v}_s + r_e \mathbf{v}_e$$

- This mathematically approximates a first-order Fréchet mean on a product manifold.
- Pass the fused outputs through a refinement block to produce a localized representation ( $\mathbf{z}_{refined}$ )

#### 6. Multi-Task Prediction Head

To avoid heavy resource footprints from maintaining twelve distinct deep prediction pipelines, implement a parameter-efficient shared architecture:

- Shared MLP Backbone: Processes the refined feature into a unified matrix ( $\mathbf{h}^{(2)} \in \mathbb{R}^{512}$ ) capturing common core behavior and intersection dynamics across all metrics.
- Lightweight Task-Specific Adaptation Adapters: Twelve individual single-layer shallow regression heads that branch from the shared backbone, projecting the output to evaluate each unique continuous metric.

#### Evaluation Strategy for the Pipeline

To accurately gauge system performance, the pipeline should be structured to jointly evaluate the 12 Continuous Target Score Metrics generated from human psychometric pairwise ground truths:

Target Outputs to Optimize

1. **Personality Metrics (The Big Five Framework):** Openness (O), Conscientiousness (C), Extraversion (E), Agreeableness (A), Neuroticism (N), and a holistic Overall Personality Index.

2. **Interview Performance Metrics:** Interview Score, Answer Score (content quality/thinking structure), Speaking Skills (tone, pace, filler words), Confidence Score, Facial Expression (engagement level), and a consolidated Overall Performance metric.

Loss Functions & Training Safeguards

Because real-world behavioral video responses feature heavily skewed distributions, non-normality, and severe outliers, your end-to-end training setup must optimize an Adaptive Multi-Component Loss Framework:

- **Huber Regression Loss ( $\delta = 1.0$ ):** Replaces standard Mean Squared Error (MSE) to prevent extreme outlier data from hijacking gradient trajectory updates.
- **Correlation Boosting Loss ( $L_{corr}$ ):** Drives Pearson correlation alignment directly into prediction behaviors.
- **Covariance Alignment Loss ( $L_{cov}$ ):** Computes Frobenius norms on prediction-target differences to force the multi-task model to preserve known human behavioral trade-offs (such as the high correlation between Extraversion and Speaking Skills).
- **Inverse Variance Loss Balancing:** Learns individual loss component weights ( $\beta_i$ ) adaptively using runtime variance updates to keep secondary task objectives from completely drowning out primary classification objectives.

3. Compare against existing systems

| Feature               | RecruitView | Internal Demo | Proposed System |
|-----------------------|-------------|---------------|-----------------|
| Video Analysis        | Yes         | Yes           | Yes             |
| Audio Analysis        | Yes         | Yes           | Yes             |
| Transcript Analysis   | Yes         | Yes           | Yes             |
| Quality Control       | Limited     | Limited       | Yes             |
| Missing Data Handling | Partial     | Unknown       | Yes             |
| Explainable Scores    | Partial     | Partial       | Yes             |
| HR Review Layer       | No          | No            | Yes             |
| Bias Monitoring       | No          | No            | Yes             |

4. Evaluation Metrics

Because human behavioral traits and interview performance scores are continuous but heavily skewed and non-normal , standard error metrics (like MSE) must be balanced with rank-based, monotonic evaluation metrics.

## Primary Evaluation Metrics

- **Spearman's Rank Correlation ( $\rho$ )** : Evaluates the monotonic relationship between the predicted scores and ground-truth utilities. It is robust to outliers and measures how well the model preserves the relative ordering of candidates.
- **Kendall's Tau-b ( $\tau_b$ )** : Computes the ordinal correspondence of pairs while explicitly accounting for ties in candidate labels.
- **Concordance Index (C-Index)**: Measures the probability that the model correctly predicts which of two randomly chosen candidate clips has a higher human assessment score.

## Secondary Error Metrics

- **Pearson Correlation ( $r$ )**: Evaluates linear correlation between predictions and labels.
- **Huber Loss / Mean Squared Error (MSE)**: Measures the absolute magnitude of pointwise deviations, with evaluation focusing on keeping average macro-errors low.

\* Final Metric Compilation = Macro-Average across all 12 target attributes

## 4.1 Validation Strategy

To prevent identity leakage, standard random k-fold cross-validation cannot be used. If frames or segments of the same individual appear in both training and validation sets, the model will overfit to individual physical traits rather than learning generalized behavioral indicators.

- **Stratified Participant-Based Split**: Splits are explicitly stratified based on an anonymized user identifier (user\_no). All responses from a single candidate are confined strictly to one partition.
- **Data Partitioning Blueprint**: \* Training Set (70%): Outlier-smoothed clips used to optimize model parameters, expert weights, and routing modules.
  - Validation Set (15%): Used for runtime monitoring, hyperparameter tuning, and triggering early stopping criteria.
  - Testing Set (15%): An completely unseen evaluation partition held back until final inference testing.
- **Early Stopping Rule**: Training tracks validation performance across epochs. Training is halted if the Macro-Averaged validation Spearman Correlation fails to improve for 5 consecutive epochs (Patience = 5).

## 4.2 Testing Procedure

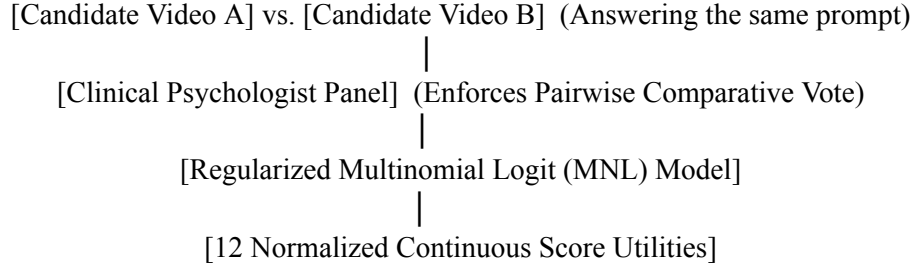
The offline inference verification execution process is organized to maintain extreme differentiability and alignment to real-world deployment:

1. **Data Ingestion & In-the-Wild Adaptation**: Unseen test video recordings undergo raw mono-audio track extraction and text transcription matching.
2. **Multimodal Encoding Block**: Tokenized text inputs pass to DeBERTa-v3-base , audio tracks to Wav2Vec2/HuBERT , and video blocks undergo 3D interpolation to compress inputs down to a standard frame-length before temporal modeling and feature extraction.
3. **Forward Pass Inference**: Features cross into the Pre-Fusion Transformer, project down into independent manifolds (Euclidean, Spherical, and Hyperbolic geometries), scale across routing weights, and integrate in the shared tangent plane.
4. **Multi-Task Metric Generation**: Fused tokens stream into the Shared MLP Backbone and fire simultaneously across the 12 specific adaptation layer heads to score the 12 psychological and interview attributes.

5. **Robust Rank Consolidation:** Pointwise outputs are converted into a global rank matrix across all candidates to verify structural alignment against global verification indices.

#### 4.3 Human Evaluation & Annotation Procedure

To establish clean, psychometrically grounded continuous score targets for model supervision and human-in-the-loop calibration, a comparative protocol is used.



#### Steps to Establish Human Ground Truth

1. **Pairwise Comparative Labeling:** Instead of using error-prone absolute rating scales (e.g., 1-10), human evaluators (clinical psychologists) review pairs of candidate responses to the exact same prompt. They select which candidate demonstrates a specific target attribute more effectively (e.g., "Who appears more confident?" or "Who exhibits better speaking skills?" ).
2. **Mathematical Utility Calibration:** Let  $j_1$  and  $j_2$  represent two video responses. The probability that  $j_1$  is preferred over  $j_2$  is mapped using a Multinomial Logit (MNL) framework:

$$\Pr\{j_1 \succ j_2\} = \frac{\exp(\theta_{j_1})}{\exp(\theta_{j_1}) + \exp(\theta_{j_2})}$$

3. **Nuclear Norm Matrix Regularization:** To handle sparse evaluation structures and exploit cross-modal score correlations, the final ground-truth matrix is solved via convex optimization with a Laplacian-induced low-rank constraint:  
This converts subjective rater interactions into reliable, continuous regression scores centered around zero.

$$\hat{\Theta} = \arg \min_{\Theta \in \Omega} \left[ -\alpha \mathcal{L}(\Theta) + \lambda \|L^{1/2} \Theta\|_* \right]$$

#### 4.4 Model Comparison Methodology

To evaluate system performance against competitive architectures, model benchmarks should be assessed under fixed baseline execution environments:

##### Selected Benchmark Baselines

Compare your proposed CRMF architecture against state-of-the-art Large Multimodal Models (LMMs) adapted for regression:

- MiniCPM-o 2.6 (8B): End-to-end small language model utilizing modality-adaptive components.
- VideoLLaMA 2.1-AV (7B): State-of-the-art model utilizing specialized spatio-temporal audio-visual compression blocks.
- Qwen2.5-Omni (7B): Advanced native audio-visual language model leveraging cross-attention fusion layers.

### Architectural Ablation Controls

To prove the structural necessity of every core module in your AI Interview system, verify performance drops by iteratively evaluating the following variants:

- **Fusion Strategy Baseline:** Replace geometric manifolds entirely with basic feature concatenation or simple weighted averaging.
- **Curvature Constraints:** Restrict geometry handling to a single space (Euclidean-only, Spherical-only, or Hyperbolic-only) to evaluate representation decay.
- **Adaptive Gating Control:** Disable the Token Router and default to uniform weights across all geometric expert fields.
- **Loss Optimization Controls:** Drop the correlation and covariance alignment loss elements, running the entire backpropagation cycle strictly on Standard MSE Loss.

### Reporting Standards

Models are ranked based on their **Macro-Averaged Performance** across all 12 dimensions , along with a **Parameter Efficiency Ratio**—computed as the model's performance relative to its total number of active, trainable parameters during fine-tuning.